

WASP-52b

R-0524

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_230723 · created 2026-07-21T02:31:05+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460148.8972 +/- 0.0075 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.214 +/- 0.062
Transit depth (Rp/Rs)^2	4.57 +/- 2.64 [%]
Orbital Inclination (inc)	85.0 +/- 2.84
Airmass coefficient 1 (a1)	1.18 +/- 0.45
Airmass coefficient 2 (a2)	-0.11 +/- 0.35
Scatter in the residuals of the lightcurve fit is	39.06 %
Best Comparison Star	None
Optimal Aperture	2.64
Optimal Annulus	5.0
Transit Duration (day)	0.07 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.214 $\pm$ 0.062 vs 0.1646 (deviation 0.8 σ)
Duration fit vs published	0.07 d vs 0.0758 d (deviation 0.22 σ)
Mid-time O–C	14.9 min
Depth SNR	3.5
Residual scatter	39.06 %

Light curve

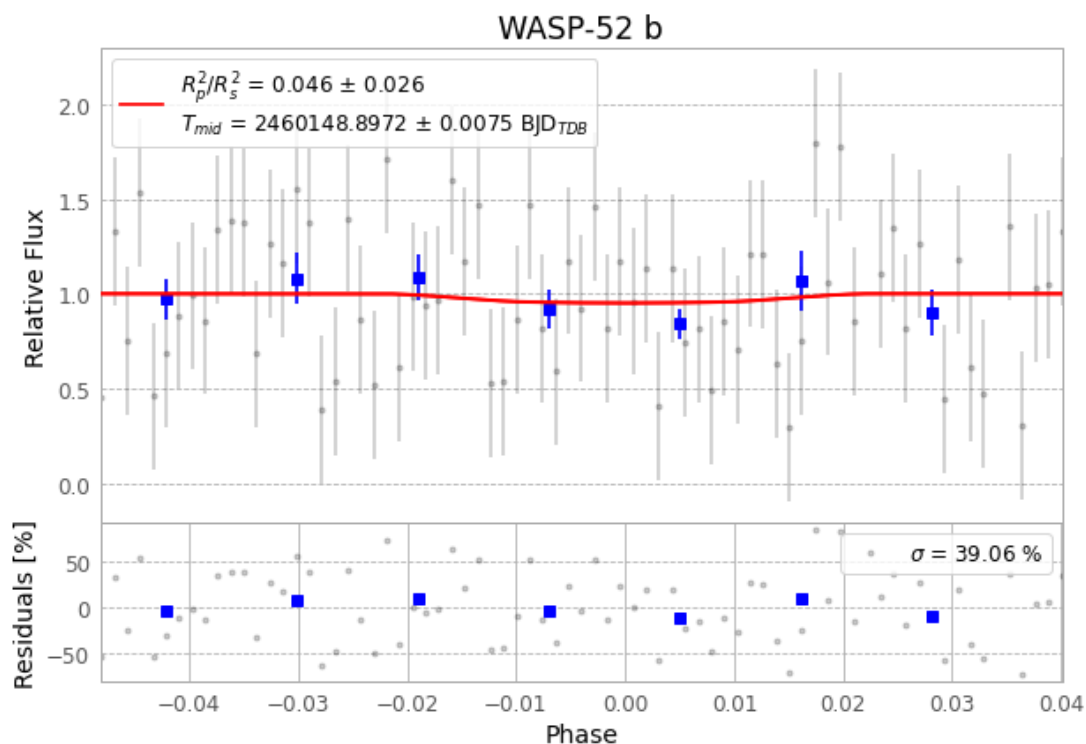


Figure 1 — FinalLightCurve_WASP-52 b_2023-07-23.png (SHA-256 5700b8a96a2ed884...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [541, 351], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9d4738b4619eff1d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 0aa9989

Data provenance

74 FITS frames (49 MB), WASP-52230723072415.FITS → WASP-52230723110615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-230723.log (a03ad7ead973...), FinalLightCurve_WASP-52 b_2023-07-23.png (5700b8a96a2e...), FinalLightCurve_WASP-52 b_2023-07-23.csv (86f1cf47295f...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-21 02:31Z.